



Original Article

Dose response relation for optic nerve atrophy at low-dose rate brachytherapy of uveal melanoma



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ABSTRACT

Background: Dose-response relationships for optic neuropathy and optic nerve atrophy after brachytherapy for posterior uveal melanoma were poorly defined from previous studies. Here, the outcome differences were analyzed in dependence on dosimetric factors, the applicator type, and tumor dependent variables.

Purpose: Primary objective was to evaluate the association of applied dose and on-set of optic nerve atrophy after brachytherapy for posterior uveal melanoma in order to allow risk estimation for new patients.

Materials and methods: This retrospective study was performed at a single high volume centre for ocular oncology. Patients receiving brachytherapy with Ruthenium-106 applicators for posterior uveal melanoma with a maximum distance between optic nerve and the nearest tumor margin of 4 optical disc diameters and follow-up with fundus photographs were included. The dose distribution at the optic nerve was reconstructed from the fundus photographs at latest follow-up and the dose-distribution of the applicator using a dedicated software. A first mask with important structural elements such as optic nerve, macula, tumor and vessels was first superimposed on the fundus photograph and adapted to the real contours. In a second step, an applicator contour mask was adapted to the radiation scar in order to calculate the dose distribution in all structures. Dose-response relations were obtained by weighted logistic regression.

Results: The maximum dose at the optic disc (OD_{max}) in this group of 109 patients ranged from 5.8 Gy – 242.2 Gy, median 48.7 Gy. Optic nerve atrophy was observed in 29 patients. Median time to radiation induced optic nerve atrophy was 18 months. Using weighted logistic regression, the dependence of optic nerve atrophy on OD_{max} was significant ($p = 0.0001$, χ^2 -test). There was a considerable interobserver variability in OD_{max} values ($p < 0.02$, signed rank test). An additional factor influencing the dose-response was the applicator type ($p = 0.0315$, χ^2 -test). The OD_{max} for a probability of optic nerve atrophy of 50 % (ED50) were 77.6 Gy $\pm$ 7.0 Gy for patients treated with notched COB applicators and 53.2 Gy $\pm$ 8.2 Gy for patients with other applicators. Including the applicator type, the area under ROC curve reached a value of 0.857 (95 %-CI: 0.793–0.921) for the logistic model with OD_{max} . The ED50 for optic nerve neuropathy, classified as grade ≥ 1 toxicity, was estimated to be 46.9 Gy $\pm$ 4.1 Gy for the maximum dose at the optic disc.

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Conclusions: Significant dose–response curves were found for optic nerve atrophy at low dose rate brachytherapy. A standard position of COB applicators was identified that allows estimation of the dose–response relation from the scleral dose of the applicator for risk estimation without fundus photographs. This larger data set enhances the knowledge of dose–response relationships for irradiation near the optic nerve.

Introduction

Uveal melanoma is the most frequent primary intraocular tumor in adults, and ocular plaque brachytherapy is the most commonly used treatment option to preserve the eye and even visual acuity in small or middle-sized tumors [1–3].

There are several potential radiation normal tissue effects which have to be evaluated [4]. Optic neuropathy is a severe diagnosis that can lead to blindness and, therefore, severely impacts on quality of life [5–8]. The optimal prevention and treatment can only be defined, if the radiation dose is known that can cause optic neuropathy after brachytherapy. Accurate diagnosis of zone and stage are essential since these determine prospective evidence-based intervention strategies [9].

In external beam radiotherapy (RT), the maximum radiation dose to the optic nerve is a known and significant factor for radiation induced optic neuropathy (RION). Thus, tolerance doses at a risk level of 5 % or 50 % at 5 years (ED5 or ED50) were estimated to maximum doses at the optic nerve of 55–60 Gy or 65–75 Gy with conventional fractionation at daily doses of 1.8–2.0 Gy, 5 fractions per week [10]. In comparison, the ED5 values for RION after single dose radiosurgery are considerably lower and were estimated to be between 8–14 Gy, while ED50 value are around 20 Gy [11–18]. Therefore, the fractionation sensitivity of the optic nerve is high. Using the linear quadratic repair model [19,20], a repair capacity of $\alpha/\beta = 0.61$ Gy can be obtained, assuming an iso-effective ED5 of a single dose of 12 Gy [11,21,18] and a total dose of 58 Gy with conventionally fractionation at 2.0 Gy per daily fraction for the optic nerve with complete recovery between daily dose fractions. This value is similar to the α/β value for spinal cord of 0.87 Gy (95 % CI: 0.54 Gy – 1.9 Gy) by Schultheiss et al. 2008 and 0.96 Gy (95 % CI: 0.24 Gy – 1.73 Gy) by Bender et al. 2012 [22,23]. The dose-rate in brachytherapy is different to that of external beam RT, and thus also the biological effect. Ruthenium-106 (Ru-106) brachytherapy belongs to low-dose-rate (LDR) radiation applications [24,25]. Experimental and clinical data indicate that the kinetics of a major proportion of > 50 % of interfraction repair in the rat spinal cord but also in the spinal cord of patients can be characterized by half times of repair of about 4 [23,26,27]. These long repair half-times indicate a rather slow repair component in serial central nervous system structures. As median treatment times with Ru-106 applicators are around 120 h, a considerable amount of repair can take place in the optic nerve papilla during treatment [16].

The symptoms of RION after brachytherapy are complex as decline of visual acuity, limitation of visual field and abnormal color vision. State of the art diagnosis of optic nerve atrophy can be made on the basis of the fundus view [28]. Patients with other causal or predisposing factors for optic neuropathy were excluded from this study. The side effects of brachytherapy on optic nerve can depend on the dose distribution at vulnerable structures and organs at risk and the latter depend on (i) location of the tumor, (ii) tumor size, (iii) height, (vi) volume, (v) radionuclide used, and its effectiveness can be influenced by the (vi) dose rate [24,29–36]. Other factors explaining toxicity include mechanical compression during surgery or bleeding, e.g., formation of blood coagulum under the calvaria [37]. These factors may affect a lot the dose distribution and thus affect the course and results of therapy. To optimize therapy and the corresponding dose profile, knowledge of individual applicator characteristics is essential [31].

A few study groups could define dose dependency of RION after

plaque brachytherapy of uveal melanoma. Precise dosimetric analysis of dose to structures of the eye are difficult as the dose rate of Ru-106 applicators decreases very rapidly both laterally and with increasing distance from the applicator surface with a steep gradient of several percent per 0.10 mm. The study group by Lommatzsch et al. analyzed the outcomes of 93 patients with juxtapapillary choroideal melanoma [38]. In that study the estimated median dose within the anterior optic nerve was 51.2 Gy (range 10.3–60.5 Gy). The probability of developing complete RION was 23 % and 53 % at 5 and 10 years, defined as complete optic atrophy [38]. The respective incidences of at sectorial optic nerve atrophy, in which one half of the papilla was still vital, in these patients were 66 % and 83 %. Oare et al. also analyzed the relationship between COMS eye plaques and side effects. There, a cohort of 52 patients was retrospectively reviewed. Clinical parameters, dosimetry, and outcomes were recorded. Forty percent of patients presented with vision decline. The doses at the optic disc were found to be above 42 Gy for these patients [39]. There are also further study groups that have seen a dose–effect relationship between RION and OD_{max}. They indicated a significantly increased risk of RION above 50 Gy disc dose [40,41].

One of the first, Heilemann et al. developed a software for estimation of applied doses [32]. They introduced a treatment planning software that allowed calculating dose-volume parameters. Even differing plaque placements were calculated with respect to the effect on tumor coverage and dose changes in organs at risk. The authors used different treatment plan optimization approaches by (i) reducing plaque size and (ii) shifting the plaque away from organs-at-risk [32]. Here, sensitive structures could be better protected with smaller applicators, as expected. Dose tolerance thresholds were not quantified.

Espensen et al. investigated the probability of complications and pathological reactions of normal tissues after brachytherapy of choroideal melanomas [24]. Ru-106 applicators were used in that institution. RION risks above 20 % were found in a broad range of the partial area of the optic disc from 10 – 100 % receiving radiation doses >50 Gy with a shallow dose response. Espensen's work used for the analysis the Plaque Simulator (version 6.5.9. EyePhysics, LLC Alamos, CA, USA), a 3D image-based irradiation planning software for ocular applicators. The Ru-106 data in this program are based on a parameterization of measured dose distributions from the manufacturer. The physical uncertainties of these basic data are more than 10 % [24]. In a prospective study, Kellner et al. analyzed the incidence of RION following Ru-106 brachytherapy of uveal melanomas [42]. Visual evoked potentials were measured before radiation and at regular follow-up intervals. Patients were grouped according to tumor location in peripheral melanomas with a central tumor margin of more than 60° to the optic nerve head and central melanomas with a tumor with contact to the optic nerve head of less than 15°. In patients with central melanomas clinical signs of RION were found in the last follow-up in 18 %, and pathological visual evoked potentials developed in 50 %. The authors concluded, that RION occurred in central tumors and not in peripheral ones, thus RION could be mainly be dependent on tumor location. A quantitative dose response relation for RION could not be provided there [42].

The exact incidence of radiation optic nerve atrophy and the exact tolerance doses for a risk of optic nerve atrophy are not yet known [43]. The focus of this work was to accurately determine the dose distribution in relation to the complication development and defining the brachytherapy dose threshold for optic nerve atrophy.

Methods

Patients

Retrospectively all data of patients with posterior uveal melanomas who underwent radiation treatment at the collaborating Departments of Ophthalmology and Radiotherapy, University Hospital Essen, Germany, from 2002 to 2015 were reviewed. Data collection was based on treatment records, radiation consultative reports, surgical reports, and photographic documentation as well as thorough follow-up. The data were matched with baseline dosimetric brachytherapy data. Standard Ru-106 eye plaques, types CCA, CCB, CCD, CCX, COB, COE, were used (Eckert & Ziegler BEBIG, Berlin, Germany). Altogether, 233 patients fulfilled the inclusion criteria of this study, i.e. a maximum distance between optic nerve and the nearest tumor margin towards the optic nerve of 4 optical disc diameters. Of these patients, 124 were excluded from the study again because the available photographic follow-up documentation did not allow for a quantitative accurate reconstruction of the dose exposure of the optic nerve from the scar produced by the applicator, in especially if the field of view of the Panoret camera photographs limited at 30° visual angle did not fully cover both the tumor bed and the optic disc sufficient for dosimetric reconstruction. Data from the remaining 109 patients were used as the basis for this study. In our department, we routinely prescribe in uveal melanoma an apex dose of 130 Gy with Ru-106 brachytherapy. Depending on the tumor height, for medium-sized tumors a dose of 130 Gy at the tumor apex, and for less prominent tumors a minimum dose of 700 Gy at the sclera was prescribed. For all included patients the corresponding dose of 85 Gy at the outer surface of a safety margin of 1.0–1.5 mm around the apex was applied [44]. In addition, the eye plaque had to cover a safety margin of 1 mm around the visible lateral tumor boundaries to ensure the application of at least 85 Gy to the complete tumor basis [45]. Juxtapapillary tumors directly adjacent to the optic disc were included within the 85 Gy isodose when the COB plaque was placed with its notch abutting the optic nerve sheet.

Optic nerve neuropathy was recorded here, when disc edema, vasoo-bilitation on the optic nerve head, haemorrhage, or formation of cotton wool foci, disc pallor or nerve fibre infarct were observed [4,46–49]. As this end point comes from fundus views without reference to vision acuity, some patients with optic nerve neuropathy were asymptomatic or had a vision $\geq 20/40$ at last follow up, this end point fulfills the criteria of an optic nerve disorder or papilla-edema of grade ≥ 1 excluding grade 5 toxicity reaction [50]. In addition, optic nerve atrophy was recorded as a more severe toxicity with optic disc pallor with sharp disc margins and a pale disc. As all patients with optic nerve atrophy on fundus view had a vision $\leq 20/200$. This end point fulfilled the criteria of a CTC-AE grade 4 toxicity [50].

In addition, time to first diagnosis of optic nerve neuropathy and atrophy during follow-up was recorded. Patients with known other causes of optic neuropathy were excluded [51].

Dosimetry and evaluation

To determine the radiation dose delivered to structures within the eye a quantitative method was developed that combined an Excel VBA program [52] to acquire the position of the Ru-106 plaque on the patient's eyeball with dose distributions across the surface of a sphere from dedicated Geant4 Monte Carlo simulations for each plaque type [53–55]. A fundus photography, covering the region around the posterior pole of the eye by a Panoret camera [56] was imported into the Excel program. In this photo the regions of interest, such as the optic disc, the fovea, blood vessels and the rim of the radiation induced scar, were manually contoured. The distances between these structures were measured. A rear projection of this image onto a spherical surface allowed for a re-calculation of these values into spatial distances on the inner surface of the eye. The uncertainties of this model due to

deviations between the sphere and the shape of the real eye were less than 1 % as long as only short distances of less than 4 mm from the center of the projection were considered for further calculations. This was valid for the present evaluation. Geant4 version 10.6 was used for the numerical simulations. For every type of evaluated eye plaque, 200 simulation procedures with 10^6 initial particles each and explicitly different setting parameters were performed. The achieved mean values and statistical uncertainties from these set of simulations were written into a look up table implemented into the Excel program. An adequate spatial calculation grid of 2° for both the polar and the azimuthal angle was essential since a disadvantageous choice of grid might cause considerable inaccuracies in the dose determination, especially for plaque types such as COB with dose gradients not only in radial direction but also along the azimuthal angle.

The activity distribution across the radioactive surface of a Ru-106 eye plaque is almost homogeneous since plaques that exceed homogeneity variations of 25 % are withdrawn from sales by the manufacturer. For most plaques, however, the inhomogeneity is significantly smaller. Thus, when the plaque is correctly fixed on the eyeball and the dose at the sclera exceeds some 400 to 500 Gy, the irradiation generates an almost regularly shaped scar reflecting the bilateral (types COB, COE) or rotational symmetry (types CCA, CCB, CCD) of the plaque, with the rims of the scar and the active plaque surface appearing concentric in fundus photography (Fig. 1a/b Supplement). This is even the case when a part of the rim region is covered by exudations. These general geometrical relations remain undisturbed even by distortions within the fundus photo due to the effects of the projection from the three-dimensional fundus view to the two-dimensional photo. Thus, the posterior rim of the scar can be used for a visually positioning of a schematic plaque model consisting of the rim of the plaque and of the active surface and projected into the fundus photo. Furthermore, the radial distances measured from the center of a fundus photo remain true to their spherical distances, especially when these distances are small, allowing for their accurate determination, and provided a suitable calibration method is available. As a basis for this calibration, the value of the perpendicular diameters of the optical disc, and the distance between optical disc and fovea were determined in each fundus photo and compared to the standard values published in literature [57–59]. In order to cope with the individual variation of these two values, the resulting calibration factor was calculated by a weighted mean value of both. The result was cross-checked by the size of the schematic plaque in the picture which was scaled by this factor, as well.

The uncertainty of this method of determination of the distances depends on the quality of the individual fundus photo. It can be estimated to be less than one-tenth of the diameter of the optic disc, that is about 0.15–0.20 mm, for the majority of cases (80 %) included in the study with sufficiently defined radiation scars. Due to the large exponential dose fall-off at the plaque rim, this results in an uncertainty of the dose value about 9 % for the majority of cases [31,60]. Since the rotational orientation of the plaque around its main axis can hardly be determined from the fundus photo – except for the notched COB or COE plaques – inhomogeneities of the activity distribution across the plaque surface cause another source of uncertainty. A survey of these inhomogeneities by means of BEBIG plaque certificates, combined with the statistical uncertainty of the numerical simulation, results in an estimated uncertainty of about 4 %. The general uncertainty of the NIST (National Institute of Standards and Technology) calibrated and traceable value for the absolute dose rate given by BEBIG is 5.5 %. The dose rate was cross-checked by measurements with a scintillation dosimeter calibrated at a strontium-90 source. This source came with a certified calibration in terms of absorbed dose rate to water, provided by the Physikalisch-Technische Bundesanstalt Braunschweig. The treatment time was recorded to the minute (placement and removal). Therefore, the combined uncertainty of the dose can be estimated to be about 11 % for the majority of the cases. Hence, this method can be considered to be quite accurate, especially when compared to the overall variation of

radiation dose around a Ru-106 eye plaque which covers several orders of magnitude within a few millimeters of range [30,31,37,61]. In about 20 % of cases included in this study, the determination of the scar was considered to be borderline acceptable, and a larger dose uncertainty up to 30 % was expected in these cases. Therefore, to estimate the dosimetric consequences of inter-observer variability of plaque positioning according to this method, two additional experienced observers independently repeated the applicator positioning of the first observer according to the fundus views in 5 patients with optic nerve atrophy at the lowest optic disc doses as estimated by the first reviewer and 13 patients without optic nerve atrophy at the highest optic disc doses by the first observer.

Statistics

The principal aim of this study was to estimate the dose–response relation of optic nerve atrophy at 60 months after therapy. As the probability of occurrence of optic nerve atrophy depends on both the radiation dose and follow-up of the patient both parameters have to be considered in the analysis. We used the method of weighted logistic regression, as proposed by Hsu et al. 2007 [62]. Here, a limited follow-up is considered by a weight factor <1 of this patient’s observation, while logistic regressions describes the dependence of the probability of optic nerve atrophy on dose and other prognostic parameters. The weight factors were derived from the Kaplan-Meier (KM) estimate of freedom from optic nerve atrophy over time (Hsu 2007) [62]. The event times of interest were the times to optic nerve atrophy and right censoring occurred for patients without optic nerve atrophy by funduscopy at last follow-up visit. The respective weight is the percentage 1-KM probability at the end of follow-up of the respective patient <60 months compared with the 1-KM probability at 60 months. Independence of the considered dose parameter and follow-up is assumed. Indeed, the Spearman’s correlation coefficient between maximum dose at optic nerve and follow-up time was -0.07 ($p = 0.47$). Kaplan Meier Analysis was performed with Life test procedure (SAS-Institute, Cary NC, USA, SAS/STAT version 15.1) [63]. The Akaike information criterion (AIC) was used to compare different dose response models. A difference between the AIC value of the best model with the smallest AIC value minus the AIC value of a candidate model smaller than -3.5 is used to reject the candidate model [64].

Spearman’s rank correlation coefficients were determined between explanatory input variables and were plotted as a correlation matrix. In addition, the correlation matrix was analyzed using hierarchical cluster analysis (Procedure VARCLUS, SAS). The clusters are created so that explanatory variables from the same cluster show a higher correlation with each other but have a low correlation with any other cluster. This procedure tries to maximize the variance that is explained by the first principal component of the variables in each cluster, summed over all the clusters. The procedure stopped, when the proportion of total variation in the data can be accounted for by the clusters components was >80 %. A dendrogram or tree diagram of hierarchical clusters was plotted.

Weighted logistic regression was performed using the SAS procedure LOGISTIC (SAS) and a stepwise parameter selection method was used. Each observation without optic nerve atrophy was given a weight according to follow-up. Observations with OD_{max} values from all three observers were all included in the analysis and the respective weight factors were divided by the number of OD_{max} values per patient.

The steepness of the logistic dose–response relation linear in dose at the dose necessary for an event probability of 50 %, the γ_{50} value, was estimated from the intercept of the logistic model [65].

The local ethics committee of the Faculty of Medicine approved the data collection and analysis for this retrospective study (study identifier: 22-10975-BO). The research was conducted in accordance with the 1964 Declaration of Helsinki. Informed consent was waived due to the retrospective data analysis with anonymized data.

Table 1

Describes the recorded and determined patient and tumor characteristics. ninety patients (82.57 %) received a notched applicator (COB, COE). Median follow-up was 25 months (6–132 months).

Patient characteristics	
Number of patients	109
Median Age (years)	66 (26–85)
Median tumor prominence before therapy [mm]	3.4 (1.0 – 7.7)
Tumors with contact to optic disc	62 / 109
tumors with ≥ 3 clock hours circumscribing optic nerve	41 / 109
tumors with ≥ 6 clock hours circumscribing optic nerve	18 / 109
Median distance between tumor and papilla of tumors with no contact to optic disc [mm]	2.0 (0.5 – 6.0)
Tumors with contact to macula	58 / 109
tumors completely overlapping the macula	34 / 109
Tumors with contact to macula and optic disc	22 / 109
Vision of the affected eye better ≥ 20/200 before therapy	94 / 109
Applicator type (CCA, CCB, CCD, CCX, COB, COE)	4 / 9 / 5 / 1 / 84 / 6
Applicator treatment time (hours)	145.7 (52.7 – 478.3)
Optic nerve maximum doses [Gy]	48.7 (5.8 – 242.2)
Optic nerve mean doses [Gy]	30.9 (3.4 – 119.0)
Optic nerve minimum doses [Gy]	19.7 (1.8 – 89.9)
Optic nerve max. dose rates (Gy/h)	0.34 (0.03 – 1.48)
Median follow-up time (months)	25 (6 – 132)
Patients that developed optic nerve atrophy	29 / 109
Patients that developed optic nerve neuropathy	48 / 109

Results

Table 1, Patient characteristics: Numbers in the rows headed by nominal parameters represent patient counts. Numbers in the rows headed by quantitative parameters represent median values. Their ranges are given in brackets.

Fig. 1a describes Kaplan-Meier curves for freedom from of optic nerve atrophy overtime. Time-dependent estimates are shown along with their point-wise 95 % confidence limits (Fig. 1a).

Twenty nine patients developed optic nerve atrophy and no one had a vision better than 20/200 at last follow-up. Twenty four of these patients had a visual acuity better than 20/200 better before therapy and all of these patients with optic nerve atrophy showed visual loss up to the last follow-up and no one had a vision better or equal than 20/400. Two of these patients received a salvage therapy before diagnosis of optic nerve atrophy which was recognized as a significant factor for the occurrence of optic nerve atrophy. One patient received a salvage brachytherapy at 35 months of follow-up and the other a central transpupillary thermotherapy at 20 months with optic nerve atrophy occurring four months later. The follow-up of these two patients was censored at the time of salvage therapy.

In addition, we analyzed whether the latency times to optic nerve atrophy in patients who developed optic nerve atrophy was depended on OD_{max}. Fig. 1b shows that patients with optic nerve atrophy and OD_{max} above the median OD_{max} of 82 Gy showed the same latency times as patients with atrophy receiving lower doses ($p = 0.95$, log rank test). The median time to optic nerve atrophy for patients who developed optic nerve atrophy was 18 months (95 %-CI: 15–24 m). The median follow-up time by the inverted Kaplan-Meier method was 30 months.

Fig. 1b shows times to optic nerve atrophy for patients developing optic nerve atrophy according to OD_{max}.

Before analyzing the dependence of optic nerve atrophy on the determined tumor, treatment, and patient related factors, the rank correlation between these explanatory factors was analyzed. OD_{min} and OD_{mean} were highly correlated with OD_{max} ($rs = 0.973$ (95 %-CI: 0.963–0.981) and $rs = 0.990$ (95 %-CI: 0.986 – 0.994)). Because of this high correlation, all three dosimetric parameters carry similar

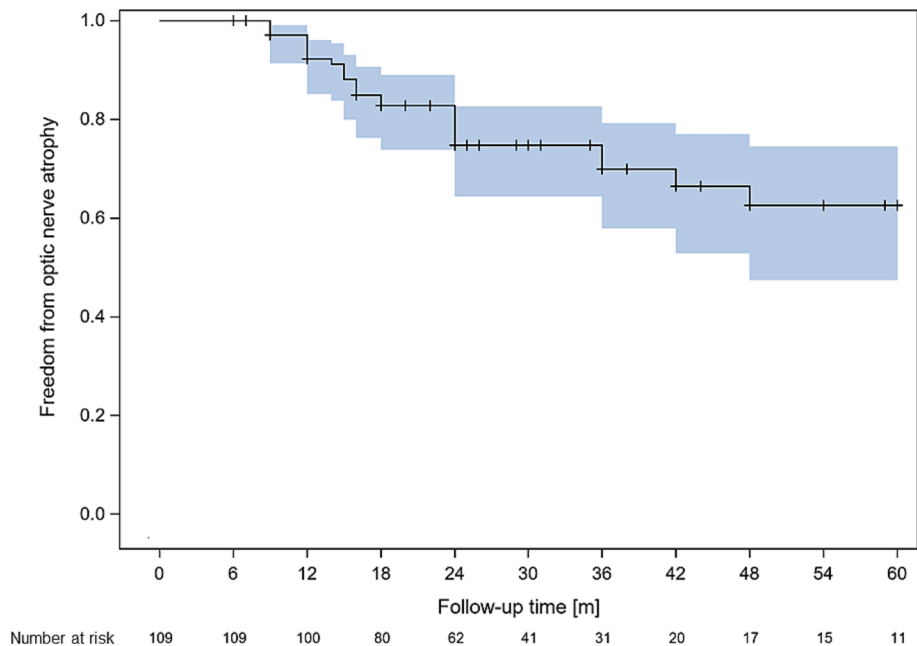


Fig. 1a. Kaplan-Meier curve for freedom from optic nerve atrophy. Time-dependent estimates are given along with their point-wise 95% confidence limits.

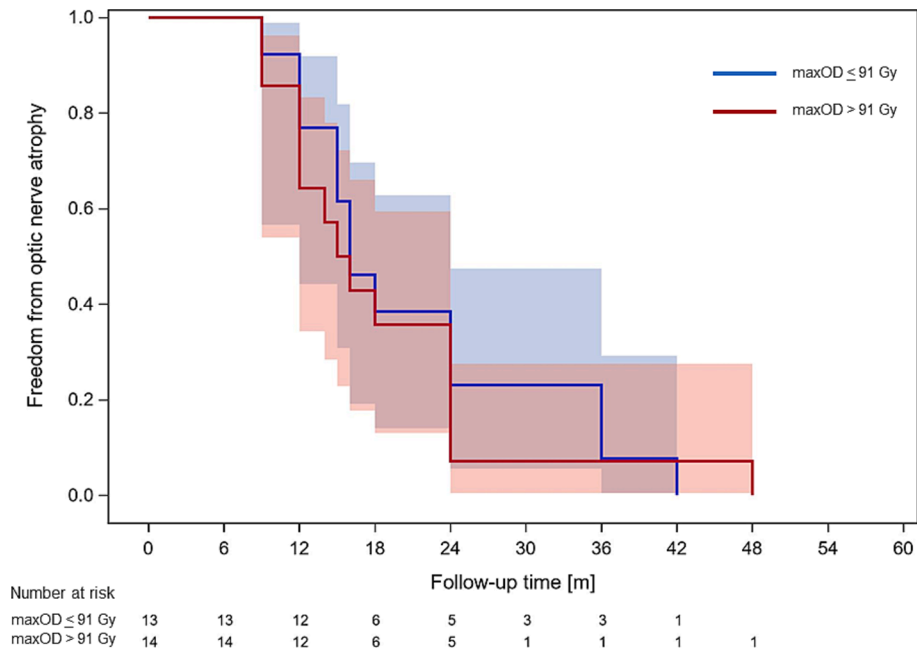


Fig. 1b. Times to optic nerve atrophy for patients developing optic nerve atrophy according to OD_{max}. Cut off was set at 91 Gy, the median OD_{max} in patients developing optic nerve atrophy. Time-dependent freedom from optic nerve atrophy estimates are shown along with their point-wise 95 % confidence limits. The curves did not differ ($p = 0.74$, log-rank test).

information. OD_{max} was chosen for further analysis as the AIC characteristics indicated better logistic regression of the optic atrophy data on OD_{max} than OD_{mean} or OD_{min}, with delta AIC values of -6.8 and -3.7 , respectively. In addition, OD_{max} is the most of reported dosimetric characteristic for optic structures and was previously used to estimate the dose-response relation for optic nerves using conventional fractionation [10].

The rank correlation between OD_{max} and all other explanatory factors was markedly lower and was largest with tumor prominence and applicator treatment time ($r_s = 0.40$ and $r_s = 0.36$). All other explanatory variables clustered farer away from OD_{max} and showed smaller

absolute values of correlation coefficients with OD_{max}.

The correlation matrix between the explanatory variables with the cluster dendrogram is shown in Table 2 together with a dendrogram as a result of cluster analysis. Treatment factors potentially associated with optic nerve atrophy were analyzed using univariable weighted logistic regression.

Spearman rank correlation between patient and treatment related exploratory factors for optic nerve atrophy after brachytherapy. Clusters that explained less than 20 % of variation by its cluster components were replaced by the cluster component with the smallest absolute correlation coefficient with OD_{max}. Therefore, the use of the notched COB was used

Table 2
Cross correlation table of variables from table used for exploratory analysis of association with optic nerve atrophy. Spearman rank correlation coefficients with absolute values larger than 0 at $p < 0.001$ were underlain with dark red and at $0.05 < p \leq 0.001$ with light red color. A dendrogram as a hierarchical binary cluster tree is given at the top of the cross correlation table. The height of the leaf nodes at which the respective cluster are joined indicate the total proportion of variance explained by the clusters at the current level of the tree. For the associations of these factors with the development of optic nerve atrophy univariable logistic regression was performed.

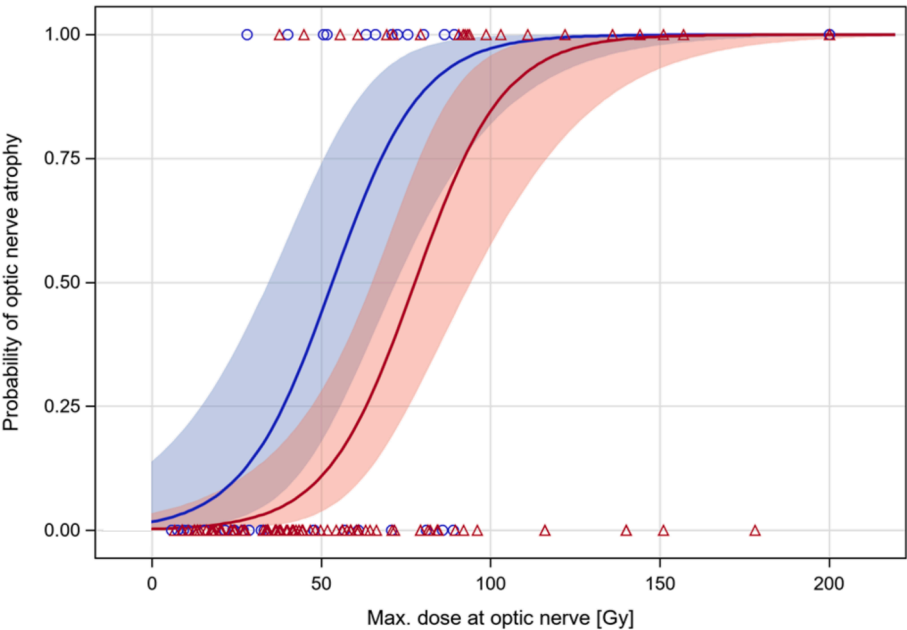
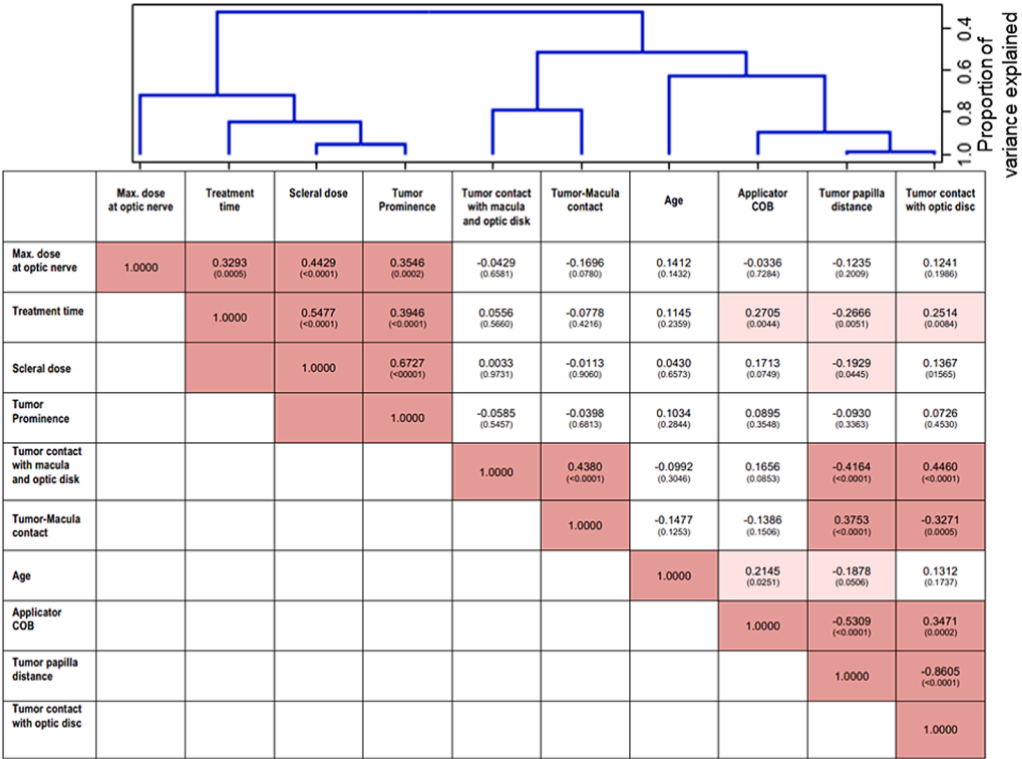


Fig. 2. Dose effect curves for optic nerve atrophy using weighted logistic regression. There was a significant dose response on OD_{max} ($p < 0.0001$, χ^2 -test). In addition, applicator type (COB applicator = yes, red curve, open triangles for observed values, versus all other applicator types, blue curve, open circles) was identified as an explanatory factor ($p = 0.0315$, χ^2 -test). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

to represent also the effect of the increasing contact of the tumor with the papilla. In addition, tumor prominence and treatment time were closely correlated.

Using stepwise selection at $\alpha = 0.05$, weighted logistic regression identified OD_{max} ($p < 0.0001$, χ^2 -test) and the applicator type (COB applicator yes or no, $p = 0.0315$, χ^2 -test) as significant explanatory factors (Fig. 2). The OD_{max} for an optic nerve atrophy risk of 50 % within 60 months of follow-up (ED50) was $77.6 \text{ Gy} \pm 7.0 \text{ Gy}$ for treatments with the COB applicator and $53.2 \text{ Gy} \pm 8.2 \text{ Gy}$ for the other applicators. We analyzed the goodness of fit of the logistic model and did not find important deviations by the Spiegelhalter goodness-of-fit test ($p = 0.89$, z-test).

In a next step we analyzed the dosimetric consequences of inter-observer variabilities in the estimation of applicator position from the fundus views for the majority of patients treated with COB applicators. Although differences between the resulting dose effect curves did not become significant ($p = 0.1106$, χ^2 -test), the ED50 values ranged from 66.7 Gy over 87.7 Gy until 111.2 Gy from observer to observer, indicating the high sensitivity of OD_{max} on small positional deviations (Fig. 3). The lowest dose at which optic nerve atrophy was observed ranged between 28–51 Gy, as estimated by three observers. Pairwise comparisons per patient showed, that OD_{max} values by the observer 2 (blue curve) and 3 (red curve) were significantly larger than that by observer 1 ($p = 0.0182$ and $p = 0.0003$, signed rank tests).

The confidence intervals of the logistic regression curves at the 5 % risk level (ED5 values) are considerably wider than that at the 50 % risk level. In addition, ED5 estimates are more affected by heterogeneities from patient to patient and of heterogeneities from observer to observer than the ED50 values so that these values are not given here.

In a next step, we analyzed whether the estimation of OD_{max} by different observers leads to a steeper and more accurate dose effect curve. This was compared with a standard COB applicator positioning geometry as the majority of patients in this study were treated with COB applicators. Assuming a standard position of the notched COB-applicator with respect to the optic nerve not considering any rotation of the COB applicator position around the axis of the optic nerve than

OD_{max} is proportional to the scleral dose. The proportionality factor leading to the same ED50 than average observed was found to be 0.07241 (Fig. 4). The resulting dose effect curve obtained from the scleral doses by the COB applicators showed no difference in slope than that by the average observer ($p = 0.7789$, χ^2 -test) demonstrating the strong relation of scleral dose and optic nerve atrophy for patients treated with COB applicators. The latter steepness could be characterized by a γ_{50} value of 1.3 ± 0.3 . Of note, the range of OD_{max} from the observers was smaller than that from the scleral doses ($p < 0.0001$, ratio mean deviations exact test alternatively Mood test). A standard geometry resulting a conversion factor between scleral dose and OD_{max} of 0.07241 is shown in Figure Supplement Fig. 1c.

The logistic regression model with OD_{max} from the average observer as a single input parameter had an area under curve (AUC) of 0.836 (95 %-CI: 0.767–0.905), which was slightly increase by inclusion of the applicator type to AUC = 0.857 (95 %-CI: 0.793–0.921).

In addition, the dose–effect relation was analyzed at the end point of optic nerve neuropathy as a grade ≥ 1 toxicity. Forty eight patients developed optic nerve neuropathy and the median time to optic nerve neuropathy $\geq$ grade 1 was 6 months (95 %-CI: 6–10 m) and as therefore shorter than median time to optic nerve atrophy. At this lower grade toxicity level of optic nerve neuropathy, the dose for a risk of 50 % within 60 months of follow-up is $46.9 \text{ Gy} \pm 4.1 \text{ Gy}$ as estimated from the OD_{max} values from all observers ($p < 0.0001$, χ^2 -test) and therefore was lower than that for optic nerve atrophy. Additional factors were not selected by the stepwise procedure. The steepness of the logistic dose response curve can be characterized by a γ_{50} of 0.83 ± 0.17 .

Discussion

Brachytherapy is one of the most effective methods treating small and medium sized uveal melanoma [31,39,43]. The primary goals are to preserve the organ and maximize remaining vision while reaching durable tumor control. Nevertheless, the radiation dose required for tumor control may exceed the tolerance of normal tissue, e.g. the optic nerve papilla (papilla nervi optici) to irradiation [3].

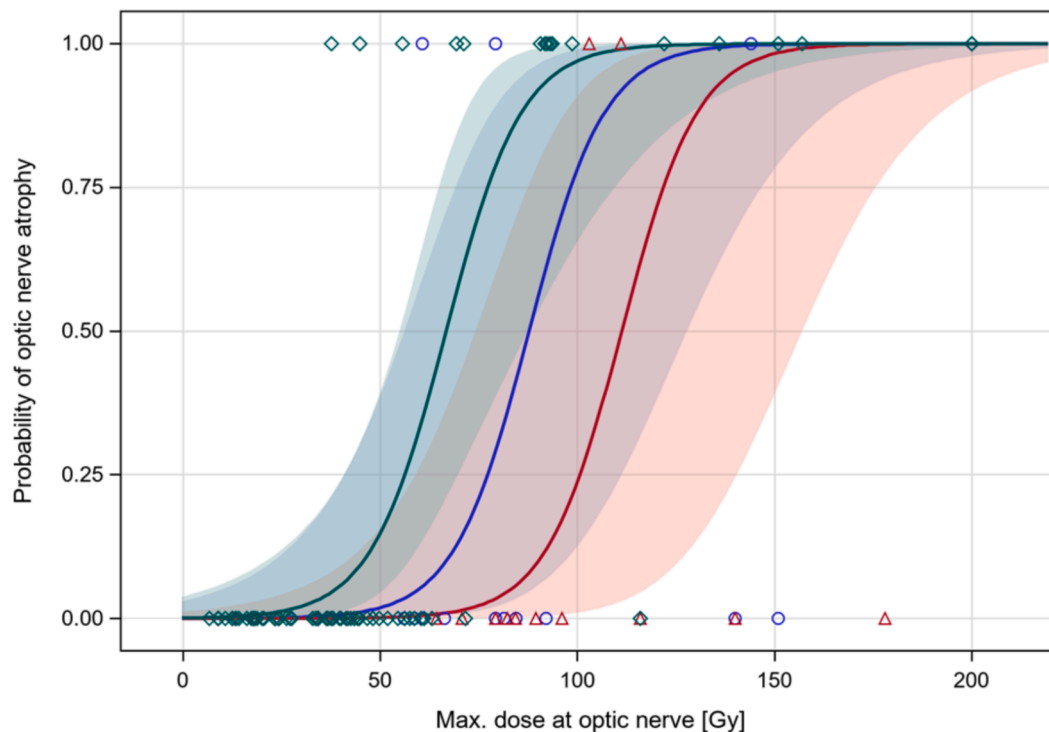


Fig. 3. Weighted logistic regression of optic nerve atrophy data on OD_{max} determined by three observers (green, blue and red curves) from patients treated with COB applicators. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

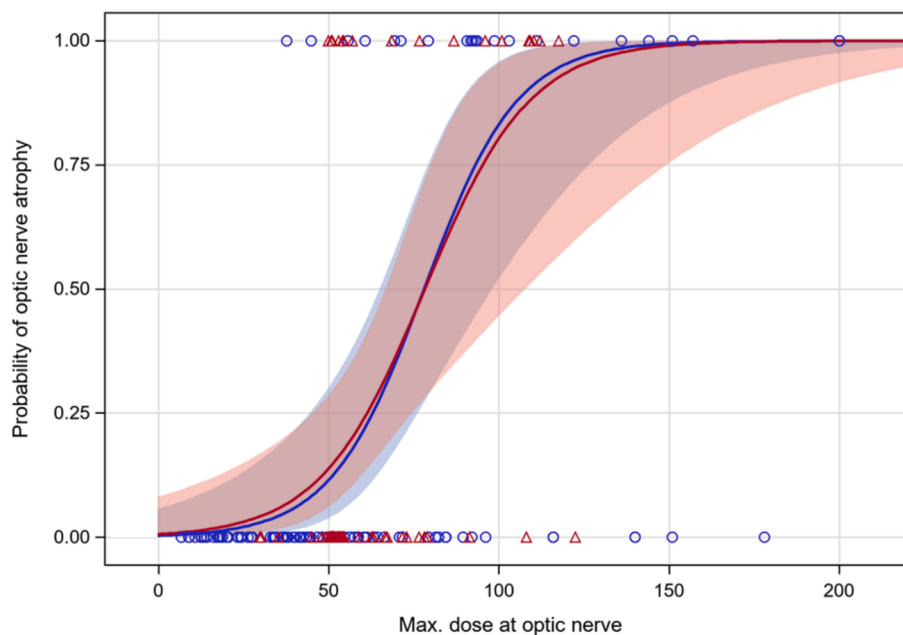


Fig. 4. Weighted logistic regression of optic nerve atrophy data on OD_{max} as determined by the average observer (blue curve and blue open circles) or from the sclera dose assuming standard applicator positioning for patients treated with COB applicators.

The consequences of irradiation occurring in this case are divided into acute and late side effects depending on the period of occurrence: (i) Side effects that occur during therapy are considered acute, (ii) effects that occur 12 weeks or later following therapy are considered late consequences e.g. caused by damage to vascular tissues. This results in decreased perfusion to the choroid leads to retinal atrophy, vascular sclerosis, and optic nerve atrophy, among others. However, Ru-106 brachytherapy is considered as one of the most gentle and effective therapies for the treatment of uveal melanoma especially due to the very steep gradients [37,61]. The aim of this work was to develop a quantitative analysis of optic nerve atrophy on the physical parameters. This will allow further refinement for the prediction of late effects of Ru-106 brachytherapy for uveal melanomas.

Normal tissue complication probability (NTCP) after plaque brachytherapy for uveal melanoma parameters were previously analyzed in a systematic review of Buonanno et al. [43]. The review included 103 studies including 21,263 UM patients and showed that Ru-106 provided very high local control outcomes but no individual patient data were analyzed. From crude incidences of optic nerve neuropathy and estimated mean doses at optic disc over the patients of the respective study using I-125 or Pd-103 applicators, they estimated the ED50 for RION to be 108 Gy at the upper confidence limit of the ED50 values from the present study using individual patient data. The ED50 for RION after brachytherapy with Ru-106 was not defined in the study by as Buonanno et al. as no study reported Ru-106 brachytherapy and overall incidences of patients with RION above 10 % [43].

The maximum dose at the papilla was given at a median dose rate of 0.42 Gy/h in this study. The ED50 for optic nerve atrophy using a notched COB applicator was $77.6 \text{ Gy} \pm 7.0 \text{ Gy}$ similar or lower than those using a conventionally fractionated schedule, that were reported to be between 65 Gy and 157 Gy [10,11,66].

While the dependence of optic nerve atrophy was highly significantly dependent on OD_{max} , marked heterogeneity was found between patients in the present study and the lowest dose at which optic nerve atrophy was observed in this study ranged between 28–51 Gy, as estimated by three observers. Heterogeneity from patient to patient and the estimated doses from observer to observer resulted in a moderate steepness of the dose–response curves characterized by a γ_{50} value of 1.3. In addition the applicator type influenced the E50 values after radiotherapy, where the

use of notched applicators resulted in a higher ED50 value. After external beam radiotherapy of patients with head and neck cancer, Moiseenko et al. found a similar γ_{50} value of 1.4 (0.8–2.2) for optic nerve neuropathy using the logistic model [66]. Milano et al. determined a γ_{50} for RION after conventionally fractionated radiotherapy [11]. Overall, a similar steepness of the dose response curve was observed here after LDR Brachytherapy as after conventionally fractionated external beam irradiation. However, dose response for an individual patient can be markedly higher than that determined for a group of patients. For normal tissues, Niemierko and Goitein noticed, that adding interindividual heterogeneity in radiation sensitivity to a fixed dose response relation will flatten the resulting sigmoid dose response [67]. The other way round, having a dose response curve to the measured data with interindividual heterogeneity and ascribing a certain amount of the observed variations to interindividual heterogeneity in radiation sensitivity, the resulting dose response for a homogeneous population of average radiosensitivity will become steeper. Similar results were obtained by Roberts et al. 1998, by incorporating intertumoral heterogeneity of radio-responsiveness in a Poisson linear quadratic model to describe tumor control probability the resulting systematic dose response curves are becoming steeper [68]. One source of interindividual heterogeneity will be the inhomogeneity of the dose distributions, delivered by brachytherapy [69,70]. Irradiation of the optic nerve with radioactive plaques is inhomogeneous. The length of the optic nerve behind the papilla exposed to high doses >50 % of the maximum dose at the optic disc is rather limited to about 2 mm [71]. The course of the optic nerve behind the papilla might slightly vary from patient to patient influencing the dose distribution.

The present study is with 29 optic nerve atrophy events from 109 patients of moderate size. Risk factors beyond maximum dose at optic nerve and applicator type were not determined here. However, the number of events in the present study is quite large in comparison to the studies included in the QUANTEC review by Mayo et al. and amounts about one fifth of the total number of events included in this review [10]. The results of this study show that evaluation of the dose distribution to critical structures is feasible and results in significant dose–response relations. The maximum dose at the optic nerve was found to be the strongest predictor for optic nerve atrophy. For notched COB applicators, it can be estimated from the scleral dose using a standard

geometry. This allows a risk assessment of optic nerve atrophy that can influence risk benefit optimizing during individualized planning.

The radio-responsiveness was even higher at the end point of radiation induced grad >1 optic nerve neuropathy diagnosed from morphological changes from the fundus with an ED50 value of 46.9 Gy $\pm$ 4.1 Gy. Radiation optic neuropathy starts with morphologic optic nerve damages such as papillitis, vaso-obliteration on the optic nerve head, edema, disc pallor or nerve fibre infarct, hemorrhage, or formation of cotton wool foci [4,72].

Several limitations of the present study have to be addressed. Shields et al. analyzed the use prophylactic anti-vascular endothelial growth factor (anti-VEGF) and intravitreal injections (bevacizumab) to minimize visual loss following plaque radiotherapy for uveal melanoma [73]. In 2019 they published results suggesting favorable reduction in radiation adverse effects on visual acuity [73]. Thus, Boldt et al. recommend a large, randomized clinical trial analyzing the fundamental mechanisms of RION in brachytherapy patients and the potential role of anti-VEGF [74]. In our cohort anti-VEGF was only applied, in case of pathological vessels, and could not be quantified [58,74,75]. Furthermore, the exact shape as well as the dimensions of the tumor were only characterized by a few numeric descriptors. In addition, the absolute dose rate value and the surface activity distribution of the applicator may have uncertainties or variations up to 10 percent points, which must also be taken into account. The largest uncertainty in OD_{max} determination was found to be the applicator positioning by the observer given the steep dose gradients at the borders of the applicator surface in this retrospective study.

Conclusion

The results of our study show that the risk of optic nerve atrophy can be predicted by maximum dose at the papilla after low-dose rate brachytherapy with Ru-106 plaques with useful precision. This allows a dose–response estimation of RION and a risk benefit optimization during individualized planning. For COB applicators a standardized position with respect to the distance to the optic nerve is appropriate for pre-treatment risk estimation.

CRediT authorship contribution statement

Maja Guberina: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Martin Stuschke:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dirk Flüh:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Leyla Jabbarli:** Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Tobias Kiefer:** Visualization, Validation, Supervision, Resources. **Eva Biewald:** Visualization, Validation, Supervision, Project administration, Methodology, Investigation. **Philipp Rating:** Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Conceptualization. **Henning Manke:** Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis. **Sami Dalbah:** Software, Project administration, Investigation, Formal analysis, Conceptualization. **Christian Hoffmann:** Validation, Software, Project administration, Investigation, Data curation. **Nika Guberina:** Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Investigation, Data curation, Conceptualization. **Christoph Pöttgen:** Validation, Supervision,

Software, Project administration, Data curation. **Miltiadis Fiorentzis:** Supervision, Software, Resources, Project administration, Methodology, Investigation. **Andreas Foerster:** Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Tobias Grunewald:** Visualization, Validation, Supervision, Resources. **Norbert Bornfeld:** Writing – review & editing, Validation, Supervision, Conceptualization. **Wolfgang Sauerwein:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration. **Nikolaos Bechrakis:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Eka-terina Sokolenko:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.radonc.2025.110775>.

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